



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 • STANDARD 6.AT.3

# Ratio Reasoning: Convert Measurements Between Systems

Lesson 3-7 • Teacher Copy

**Name:** Type your name

**Date:** Today's date

**Class:** Class period

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can use ratio reasoning to convert measurements between the customary and metric systems.

**Language Objective:** I can explain a conversion between systems using the words conversion factor, approximately, customary, and metric.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Convert Measurements Between Systems

Conversion factor

Customary system

Metric system

Approximately

Ratio table



Tap any word to see what it means and a picture.

1 Rewriting a customary measurement as a metric one gives an

answer, not an exact one.

2 The ratio that trades one unit for another is a

3 Inches, pounds and gallons belong to the

system.

4 Centimeters, grams and liters belong to the

system, built on tens.

5 Close to but not exactly equal is

— written with  $\approx$ .

6 A table whose every column holds the same comparison is a

  .

## Write About the Math

**Is 100 kilometers per hour faster or slower than 100 miles per hour, and how do you know?**

*¿100 kilómetros por hora es más rápido o más lento que 100 millas por hora, y cómo lo sabes?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Convert Measurements Between Systems**  
*Convertir medidas entre sistemas*

☐ **Conversion factor**  
*Factor de conversión*

☐ **Customary system**  
*Sistema usual*

☐ **Metric system**  
*Sistema métrico*

☐ **Approximately**  
*Aproximadamente*

**Model:** A mile is longer than a kilometer ( $1 \text{ km} \approx 0.6 \text{ mi}$ ), so 5 miles is the longer race even though the numbers match. — Students reason about UNIT SIZE before touching arithmetic — same number, different units, different lengths.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- The conversion factor is  $1 \text{ km} \approx \underline{\hspace{1cm}}$  mi. Scaling that up, 100 kilometers  $\approx \underline{\hspace{1cm}}$  miles, so the sign means about  $\underline{\hspace{1cm}}$  miles per hour. That is  $\underline{\hspace{1cm}}$  than 100 miles per hour, because in one hour you would travel  $\underline{\hspace{1cm}}$  distance.
  - My answer is  $\underline{\hspace{1cm}}$  because  $\underline{\hspace{1cm}}$ .  
*Mi respuesta es  $\underline{\hspace{1cm}}$  porque  $\underline{\hspace{1cm}}$ .*
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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is  $\underline{\hspace{1cm}}$ .

*Mi afirmación es \_\_\_\_.*

- **My evidence is** \_\_\_\_, **so** \_\_\_\_.

*Mi evidencia es \_\_\_\_, entonces \_\_\_\_.*

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## Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

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## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Convert Measurements Between Systems
2. **Notes 2:** Conversion factor
3. **Notes 3:** Customary system
4. **Notes 4:** Metric system
5. **Notes 5:** Approximately
6. **Notes 6:** Ratio table
7. **Watch for this mistake:** *A common mistake converting between systems is comparing the bare numbers without converting at all — deciding that 5 kilometers and 5 miles are the same distance because both say 5. They are not: 5 kilometers  $\approx$  3 miles. A second mistake is writing = instead of  $\approx$ . Conversion factors between systems are rounded (1 mile is about 1.609344 kilometers), so the result is approximate no matter how carefully you multiply.*
8. **Try It 1:** — *Each pair is a conversion factor between the two systems, and each one is approximate.*
9. **Try It 2:** A. The conversion factors are rounded, so results are approximate — *One mile is about 1.609344 kilometers — a decimal that never ends neatly. Any usable factor is rounded, so the answer is approximate.*